INCH-POUND

MIL-B-44394(GL)
29 December 1989

MILITARY SPECIFICATION

BEEF SNACKS-CURED SAUSAGE, OPERATIONAL RATION COMPONENT

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers beef snacks-cured sausage for use by the Department of Defense as a component of operational rations.

1.2 Classification. The beef snacks shall be of the following types as specified (see 6.1).

Type I - Plain
Type II - Spiced

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN SPICE AND TRADE ASSOCIATION (ASTA)

Official Analytical Methods of the American Spice and Trade Association

(Application for copies should be addressed to the American Spice and Trade Association, Incorporated, 580 Sylvan Avenue, Englewood Cliffs, NJ 07632.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef, ground, special. The ground beef shall conform to Institutional Meat Purchase Specifications (IMPS) Item Number 137, except that the ground beef shall have a final grind through a plate having holes measuring 3/8 inch in diameter and shall have a total fat content not to exceed 8.0 percent (see 6.4). Handling and storage of the beef shall comply with 3.2.1.1 or 3.2.1.2.

3.2.1.1 Beef, ground, chilled. Ground beef received in the chilled state shall not have been previously frozen and shall have been held at an internal temperature of 28° to 40°F for a period not to exceed 72 hours following initial chilling and prior to product preparation.

3.2.1.2 Beef, ground, frozen. Ground beef received in the frozen state shall have been held at a temperature below 0°F for a period not to exceed 120 days following initial freezing and prior to product preparation.

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3.2.2 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall possess a fragrant, clove-like aroma, a strongly aromatic, pungent, clove-like flavor, and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.3 Ascorbic acid, encapsulated. Ascorbic acid shall be encapsulated with partially hydrogenated soybean and cottonseed oil with a melting point between 124° and 130°F. The encapsulated ascorbic acid shall be white to off-white free flowing granules such that only a trace remains on a U.S. Standard No. 14 sieve. The ascorbic acid content shall be between 68 and 72 percent and the ascorbic acid shall comply with the Food Chemicals Codex (see 6.5).

3.2.4 Coriander, ground. Ground coriander shall be prepared from the dried fruit of Coriandrum sativum L. and shall possess a slight fragrant odor, a pleasant aromatic taste, and a yellowish-brown to tan color. The ground coriander shall contain not less than 0.3 mL of volatile oil per 100 grams of ground coriander and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.5 Fennel, ground. Ground fennel shall be derived from the ripe fruit of Foeniculum vulgare Mill. The ground fennel shall be light green-brown to yellow-brown in color and shall possess a pleasant aromatic odor with a flavor resembling that of anise. The fennel shall contain not less than 1.0 mL of volatile oil per 100 grams of ground fennel and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.2.6 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.7 Lactic acid, encapsulated. The lactic acid shall be encapsulated with partially hydrogenated palm oil with a melting point of 136° to 143°F. The encapsulated lactic acid shall be white, free flowing and granular such that 0 percent remains on a U.S. Standard No. 10 sieve. The lactic acid content shall be not less than 52 percent with an actual free lactic acid content of between 30 and 34 percent (see 6.5). The lactic acid shall comply with the Food Chemicals Codex.

3.2.8 Paprika, ground. Ground paprika shall be Spanish paprika (Capsicum annuum L.) and shall possess a bright orange to red color with an extractable color value of not less than 110 American Spice and Trade Association color units. The ground paprika shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

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3.2.9. Pepper, black, ground. Ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot, biting, pungent flavor, and a light gray to speckled black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and be of such size that 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.10 Pepper, red, ground. Ground red pepper shall be derived from red, ripe fruit of Capsicum fructesens L. and shall possess the characteristic yellowish-red to red color. The Scoville Pungency Value shall be not less than 30,000 units. The red pepper shall be uniformly ground to allow a minimum of 95 percent, by weight, to pass through a U.S. Standard No. 40 sieve, and not less than 95 percent, by weight, to be retained on a U.S. Standard No. 60 sieve.

3.2.11 Salt. Salt shall be white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.12 Sodium ascorbate. Sodium ascorbate shall comply with the Food Chemicals Codex.

3.2.13 Sodium nitrite. Sodium nitrite shall comply with the Food Chemicals Codex.

3.2.14 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof.

3.2.15 Water. Water shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of beef snacks. Type I and type II beef snacks shall be formulated and prepared as follows:

3.3.1.1 Type I formulation. Type I beef snacks shall be formulated as follows:

Ingredient	Percent by weight
Beef, ground	97.695
Salt	2.090
Pepper, black, ground	0.100
Pepper, red, ground	0.050
Sodium ascorbate	0.050
Sodium nitrite 1/	0.015

1/ Sodium nitrite shall be of sufficient strength to ensure a complete cure without exceeding 150 parts per million of nitrite in the cured beef during preparation.

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3.3.1.2 Type II formulation. Type II beef snacks shall be formulated as follows:

Ingredient	Percent by weight
Beef, ground	96.1
Salt/nitrite mix	2.1
Spice/lactic acid mix	1.8
Salt/nitrite mix	
Salt	99.3
Sodium nitrite 1/	0.7
Spice/lactic acid mix	
Paprika, ground	40.0
Sugar, white, granulated	30.6
Lactic acid, encapsulated	8.8
Pepper, black, ground	5.5
Ascorbic acid, encapsulated	3.9
Fennel, ground	3.3
Pepper, red, ground	2.8
Garlic powder	2.8
Coriander, ground	1.7
Allspice, ground	0.6

1/ Sodium nitrite shall be of sufficient strength to ensure a complete cure without exceeding 150 parts per million of nitrite in the cured beef during preparation.

3.3.1.3 Preparation of type I and type II beef snacks. Type I and type II beef snacks shall be prepared as follows:

a. The ground beef shall be mixed with the salt and nitrite until just blended. The spices and sodium ascorbate (type I) or spice/lactic acid mix (type II) shall then be added and the total mixed until tacky. Small amounts of water may be used to dissolve cure ingredients prior to addition or as necessary for forming. Beef snack mixture shall not exceed 40°F after mixing.

b. The beef snack mixture shall be held for sufficient time (not to exceed 24 hours) in the temperature range of 28° to 40°F to ensure a complete cure (uniform gray color) prior to forming.

3.3.2 Forming and slicing. The cured beef snack mixture shall be immediately formed into slices as follows and to ensure compliance with finished product requirements (see 3.5):

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a. The cured beef snack mixture shall be either mechanically stuffed into casings or formed into loaves and tempered to a uniform temperature between 10° and 25°F. The product shall be mechanically sliced to a thickness between 1/8 and 3/16 inch.

b. Alternatively, the cured beef snack mixture shall be directly mechanically formed into slices between 1/8 and 3/16 inch thick with a Formax or equivalent equipment.

c. The cured and formed beef snacks shall be maintained within the temperature range of 28° to 40°F for not more 48 hours prior to placement in the smokehouse.

3.3.3 Smoking/cooking. The cured and formed beef snack slices shall be placed in the smokehouse and dried such as to comply with the requirements of 3.5. Natural smoke from hickory, post oak, maple, or a combination of these woods shall be applied for not less than 1 hour at the beginning of the cook cycle.

3.3.4 Cooling. After removal of beef snacks from the smokehouse, they shall be cooled to ambient temperature before packaging. If beef snacks are held longer than 24 hours after cooking, they shall be flushed with and held under nitrogen until unit packed; however, product shall be unit packed within 72 hours after removal from smokehouse.

3.4 Pouch filling and sealing. The beef snacks shall be packaged in accordance with 5.1.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The net weight of the pouch shall be not less than 32.0 grams nor greater than 39.0 grams.

e. Dried beef snack slices shall be rectangular between 1.8 and 2.2 inches wide and between 2.6 and 3.0 inches long.

f. Water activity shall be between 0.65 and 0.75 when measured at 25°C.

g. Fat content shall be between 10.0 and 20.0 percent.

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h. Beef snacks shall be a reddish brown color.

i. Beef snacks shall have a smoky odor and a salty, smoky flavor.

j. Beef snacks shall have a firm, chewy texture.

k. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

l. The E. coli count shall be less than 3 per gram (no positive tubes in the standard three tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample in palatability and overall appearance.

3.6 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated pouch material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraph	Test procedure
Laminated material identification and construction	5.1.1.1	Laboratory evaluation.
Polyolefin and polyester film thickness	5.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	5.1.1.1	As specified in QQ-A-1876
Color	5.1.1.1	Visual evaluation

4.5.1.3 Unfilled pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to the requirements specified in 5.1.1.1. When deemed necessary by the USDA, testing of the unfilled preformed pouches for seal strength and integrity and air tightness of manufacturer's seals shall be as specified in 4.5.1.3.1 and 4.5.1.3.2.

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4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.3.2 Unfilled preformed pouch integrity and air tightness of manufacturer's seals testing. Pouch integrity and air tightness of seals shall be determined by placing rigid material similar in size and shape to the beef snack slices into the pouch and sealing under a vacuum of not less than 25 inches of mercury. Not less than 96 hours after vacuum sealing, the pouch shall be examined for vacuum loss in accordance with the procedure in 4.5.4. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Pouch size not as specified 2/
102		Not heat sealed as specified

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TABLE I. Filled and sealed pouch defects 1/ (cont'd)

Category		Defect
Major	Minor	
103		Not clean 3/
104		Tear, hole, or open seal
105		Pouch has foreign odor
106		Evidence of delamination 4/
107		Seal widths not as specified
108		Closure seal located less than 1/8 inch above tear notch
109		Presence of foldover wrinkles, occluded matter, and entrapped moisture or grease that reduces the effective closure seal to less than 1/8 inch 5/
110		Required labeling or marking missing, incorrect illegible, or that smudges
111		Dimensions of stacked slices greater than specified (see 5.1.1.2)
	201	Tear notch missing
	202	Tear notch not located as specified
	203	Depth of tear notch not as specified

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Measurement may be accomplished on unfilled pouches being used for current production lot.

3/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean dry cloth.

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b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

4/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between the thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

5/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/8 inch wide from side seal to side seal that produces a hermetically sealed pouch.

4.5.4 Pouch vacuum examination. Not less than 96 hours after filling and sealing, the pouches shall be visually examined for conformance to the vacuum requirements in 5.1.1.2. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 32.0 grams or greater than 39.0 grams shall be classified as a minor defect. The lot size shall be expressed in pouches. The results shall be reported to the nearest 0.1 gram. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.5.6 Product examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 2.5 for minor defects.

TABLE II. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Dimensions of slices not as specified
102	Color not as specified
103	Odor and flavor not as specified
104	Not a firm, chewy texture
201	Number of slices not as specified

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent).

3/ Determine once per sample unit.

4.5.7 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 inch in width and to a length suitable for proper mounting. Three adjacent specimens shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.2 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-1.

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4.5.8 Water activity (Aw) testing: The finished product shall be tested for water activity in an AMS laboratory in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Vegetable Products, Processed; Method: Water Activity-Official First Action, using a hygrodynamic hygrometer system or an equivalent instrument. The sample unit shall be the contents of one filled and sealed pouch and the product shall not be ground prior to analysis. Water activity measurement shall be performed at 25°C. The results shall be reported to the nearest 0.01 units. Any result not conforming to the requirement in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The inspection level shall be S-2.

4.5.9 Fat content testing. The finished product shall be tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products. Results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat requirement in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-2.

4.5.10 Microbiological testing. Five filled and sealed pouches shall be selected at random from the lot regardless of lot size. The pouches shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Microbiological Methods; Section: Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each slice and allowed to stand for 15 minutes before blending the sample. Continue with the procedure as directed in the Official First Action. Any result not conforming to the microbiological requirements in 3.5 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.11 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Cushioning material missing or not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Not required number of pouches packed in a container.
Pouches not packed flat in layers.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The product shall be unit packed in accordance 5.1.1.1 and 5.1.1.2

5.1.1.1 Pouch material and construction. The preformed pouch shall be fabricated from a multi-ply material, constructed from inside to outside of 0.003 to 0.004 inch thick polyolefin, 0.00035 to 0.0007 inch thick aluminum foil, and 0.0005 inch thick polyester. Color shall be applied to either the exterior or interior surfaces of the polyester lamina. If applied externally, the color shall be in the range of 34079 through 34087 or shall be 10045 or 30045 through 30118, excluding color 30109 of FED-STD-595. If applied internally, the color shall be in the range of 24052 through 24087. The color shall be applied completely and uniformly to the polyester lamina. The color application process shall not adversely affect the material performance. The material shall not transfer any foreign odor or flavor to the packaged product. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into pouches. The preformed pouch shall be a flat style pouch having inside dimensions of 2-3/4 inches wide by 4-3/4 inches long (± 1/16 inch), with the 2-3/4 inch dimension being the opening of the pouch. The pouch shall be made by heat sealing three edges with 1/4 inch (± 1/16 inch) wide seals. The heat seals shall be made in a manner that will ensure hermetic seals. The pouch shall maintain its integrity and air tightness of the side and bottom seals when tested as specified in 4.5.1.3.2. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.1.3.1. A tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the pouch, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch shall be provided with an extended lip, extending 1/8 inch (± 1/16 inch) to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended lip.

5.1.1.2 Filling and sealing of unit pack pouches. Between 32.0 grams and 39.0 grams of product slices (4 to 6 slices) shall be placed into a preformed pouch as specified in 5.1.1.1. The filled pouch shall then be hermetically sealed by evacuating the air within a sealing chamber under a vacuum of not less than 25 inches of mercury, and applying a continuous heat seal not less than 1/8 inch wide. The closure seal shall be free of foldover wrinkles, delamination, occluded matter, and entrapped moisture or grease. The closure seal shall be located not less than 1/8 inch above the tear notch. The average closure seal strength shall be not less than 6 pounds per inch and no individual test specimen shall be less than 5 pounds per inch when tested as specified in 4.5.7. After sealing, the beef slices shall remain stacked such that the maximum dimensions of the stacked slices shall not exceed 2.5 inches in width by 3.3 inches in length. The filled and sealed pouches shall show no loss of vacuum when examined as specified in 4.5.4.

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5.2 Packing for shipment to ration assembler. Four hundred and fifty pouches of product, preserved as specified in 5.1.1, shall be packed flat in layers of 50 pouches per layer. The pouches shall be packed in a snug-fitting fiberboard box conforming to style RSC-L, type CF, variety SW, Grade 275, class domestic of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick, conforming to type I of PPP-C-1797 or cushioning material conforming to type VII, class 1 of PPP-C-1752 shall be placed between each of the layers and at the bottom and top of the box to minimize movement of the individual packs.

5.3 Labeling and marking.

5.3.1 Unit pack pouches. Unit pack pouches shall be labeled in a permanent, contrasting color directly on the packaging material in a manner that does not damage the pouches, with a food compatible ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water-resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water-insoluble, homogenous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low temperatures. Dimensions of the label shall not be larger than the dimensions of the surface of the product where the label is applied on the package. Ink and adhesive used must be compatible for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 1/16 inch high. The following information must be included:

Item Name
Date of production/lot number (Julian date) 1/
USDA Inspection Marking
Name of contractor

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code.

5.3.2 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Fat content. It has been found that raw ground meat having a fat content of between 5.0 and 8.0 percent has produced a product which meets finished product fat requirements.

6.5 Encapsulated acids. It has been found that Durkote (R) Vitamin C 125-70 and Durkote (R) Lactic Acid 135-50 meet the requirements of 3.2.3 and 3.2.7, respectively, and perform satisfactorily in this product. Other encapsulated acid products may be equally acceptable however melting points shall not be below 125°F and formula adjustments may be required to adjust for free acid content.

6.6 Subject term (key word) listing.

Combat field feeding
Food processing
Ration Lightweight, 30-Day

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Custodian:
Army - GL

Review Activities:
Army - MD, QM
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:
Army - GL

(Project 8940-A669)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44394(GL)	Beef Snacks-Cured Sausage, Operational Ration Component

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 02 MAR

PREVIOUS EDITION IS OBSOLETE.

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